

# Business Forecasting Project 2 Final Report

## Sales Forecasting Framework for Retail Operations

Business Forecasting (WIHN0032)  
Prof. Dr. Dr. h.c. Gudrun P. Kiesmüller  
Dr. Dominik Proschmann

Yu-Ching Lin, Xueying Peng

### 1. Introduction

This project aims to generate 6 week sales forecasts for 676 stores of a European drugstore chain. We first performed data preprocessing and exploratory data analysis (EDA) to understand sales patterns and identify important influencing factors. Statistical benchmark models were then established to provide baseline performance for comparison. Finally, a global Random Forest forecasting framework was developed by integrating historical sales, engineered time series features, store metadata, and operational variables. Since future customer counts are unavailable but strongly related to sales, a customer forecasting model was first constructed, and its predictions were incorporated into the final sales forecasting model.

The forecasting performance of the proposed framework is evaluated and compared with benchmark models to identify an accurate and practical solution for large scale retail forecasting.

### 2. Data Understanding

Three datasets were provided for this project: sales, future, and metadata. Together, these datasets contain historical sales records and future operational information.

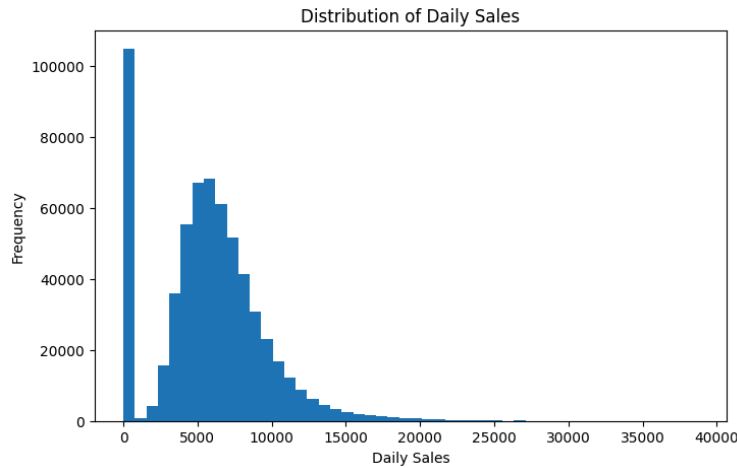
The historical sales dataset consists of 624,624 daily observations collected from 676 stores, covering the period from 07.01.2013 to 19.07.2015. Each record contains daily sales, customer counts, store operating status, promotional activities, and holiday information. The future values dataset provides known future operational variables for the forecasting period from 20.07.2015 to 17.09.2015, while the target variable (sales) is intentionally unavailable because it represents the forecasting objective. The metadata dataset contains static information, including store type, assortment level, and competition distance.

Although future data contains 60 future dates, the project requires a 6 week horizon. Therefore, only the first 42 dates were used for the final forecast and business analysis. These three datasets complement each other by combining historical sales record, future operational conditions and some store characteristics, providing the necessary information for building a 6 week sales forecasting framework.

### 3. Exploratory Data Analysis

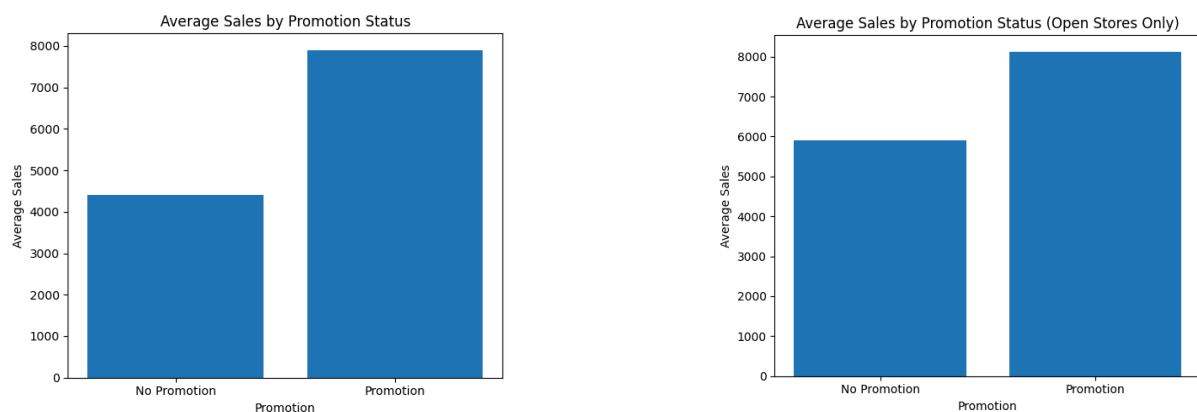
To understand the dataset's characteristics and identify key drivers of sales variance, an Exploratory Data Analysis (EDA) was initially conducted. This analysis serves as the foundation for subsequent model construction and feature engineering.

First, an examination of the daily sales distribution across all stores revealed pronounced differences, indicating that each store exhibits distinct sales patterns and demand profiles. Therefore, when developing predictive models, it is essential to include different characteristics of all stores alongside historical sales data, rather than relying only on aggregate or average historical behaviors.



**Fig.1 Distribution of Daily Sales**

Second, the impact of promotional activities on sales was evaluated. The results indicated that the average sales during promotional periods were significantly higher than those during no promotional periods. This demonstrates that promotions are a critical external factor driving demand, thereby justifying the retention of the promo variable as an important predictor in subsequent modeling stages.



**Fig. 2 Average Sales by Promotion Status**

Furthermore, the influence of holidays on sales was analyzed. During national and festival holidays, sales in some stores dropped close to zero, primarily driven by scheduled store closures. Consequently, state holiday and open serve as effective indicators for capturing sudden shifts in demand, offering substantial predictive value for sales forecasting.

Given that customer traffic directly reflects store demand, this study also investigated the relationship between the number of customers and total sales. The analysis revealed a strong positive correlation between the two metrics (correlation = 0.8822), establishing customer traffic as a primary driver of sales performance. Since future customer counts are

unavailable during the actual forecasting horizon, introducing a predicted customer variable into the sales model risks cascading forecasting errors from the customer model into the final sales predictions. To mitigate this risk, the baseline benchmark phase excluded the customer variable. Nonetheless, during the Random Forest modeling phase, we made two separate versions: one incorporating and one omitting the customer feature, to systematically evaluate its impact and examine the resulting differences.

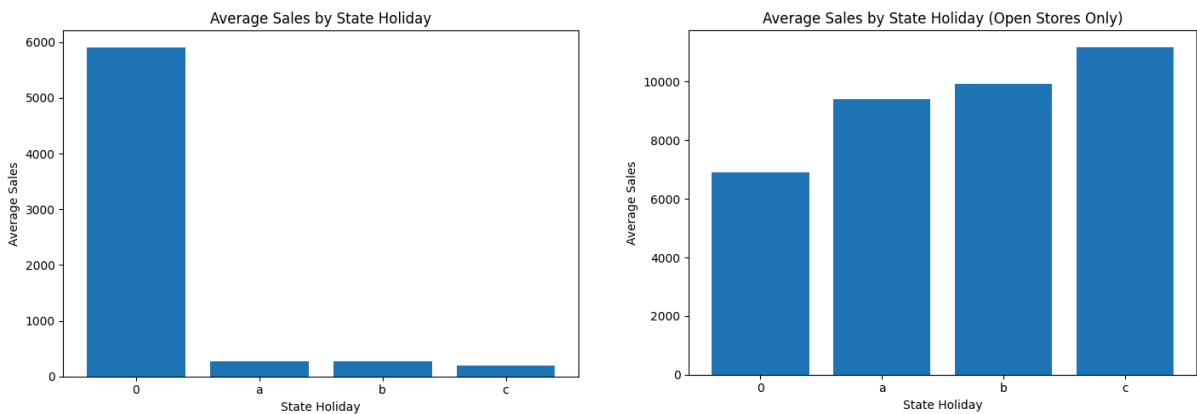


Fig.3 Average Sales by State Holiday

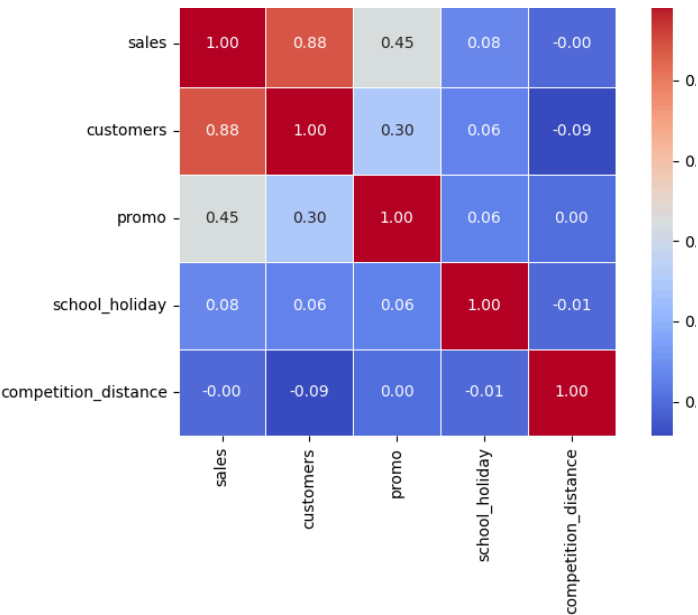


Fig.4 Correlation Matrix of Important Variable and Sales

We also observed the temporal dynamics of daily sales, which uncovered a distinct weekly seasonality characterized by stable demand fluctuations across different days of the week. To capture these time dependent patterns, various time series features were integrated into the subsequent models, including lag features, rolling statistics, and calendar features.

In summary, the EDA demonstrates that sales are not merely a function of past sales volumes; rather, they are jointly shaped by promotional activities, holiday effects, store operational statuses, customer demand, temporal cycles, and inherent specific store

characteristics. These analytical insights directly inform the design of the subsequent feature engineering pipeline and forecasting models.

## **4. Data Preparation**

### **4.1 Data Cleaning**

Before model development, the three raw datasets were examined for missing values, duplicate observations, data type inconsistencies, and store coverage. The historical sales dataset contained 624,624 observations for 676 stores and had no missing values in any of its original variables and duplicated rows. The future dataset contained 40,560 observations, corresponding to 60 future dates for all 676 stores. It also contained no duplicates. The metadata file contained one record for each of the 676 stores, with no duplicates.

Three types of missing values were identified. First, the open variable contained 11 missing observations in the future dataset. All 11 records belonged to store\_379 and occurred on weekdays or Saturdays. Inspection of the store's recent historical operating pattern showed that it was normally open on these days and closed on Sundays. Therefore, these 11 values were imputed as 1, indicating that the store would be open.

Second, competition distance contained one missing value in the metadata. The observed values ranged from 20 to 75,860 metres and were positively skewed, with a median of 2,640 metres. Because only one value was missing and the median is less sensitive to extreme distances than the mean, the missing value was replaced with the median.

Third, all 40,560 values in the future customers column were missing. These values were not treated as ordinary errors because future customer numbers were genuinely unavailable at the time of forecasting. Customer demand was handled separately by developing a customer forecasting model, whose predictions were later used as inputs to the final sales model (in one of Random Forest Model).

Date variables were converted to datetime format, while categorical variables such as store\_id, state\_holiday, store\_type, and assortment were assigned appropriate categorical data types. Store metadata were then merged into both the historical and future datasets using store\_id. After cleaning and merging, no missing values remained except for the intentionally unavailable future customer counts.

### **4.2 Feature Engineering**

Preparing for Random Forest Model, we did some serious data cleaning and feature engineering. First, we sorted everything strictly by store ID and date to keep the timeline intact, which ensures our time series features are accurate. We also renamed the key columns, changing store id to unique id, date to ds, and sales to y, to match what the ML forecasting package expects. For the categorical variables, we used mapping dictionaries to turn them into integers, state\_holiday values (0, a, b, c) became 0, 1, 2, 3; store\_type (a, b, c, d) became 0, 1, 2, 3; and assortment (a, b, c) became 0, 1, 2.

When it came to setting up the time and contextual features, we used a mix of manual engineering and package tools to capture different angles of the timeline. We manually created features for quarter week of year, and also multiplied the promo and weekend

variables to create an interaction feature called `promo_weekend`. Core calendar features like month were left to the forecasting package to extract automatically during training.

To really capture the trends and historical momentum in the time series, we set up key lag features for the final sales model spanning a bit over four weeks, including 1, 7, 14, 21, and 28 day lags. At the same time, we built two rolling average features: a 4 day rolling mean based on lag 7 (`RollingMean(window_size=4)`), and a 2 day rolling mean based on lag 28 (`RollingMean(window_size=2)`). This helps the model balance short term ups and downs with smoother long term baselines.

We also tried integrating historical actual customers with future customer predictions. From there, we generated extra features like `customer_promo` (predicted customers during promos), `customer_weekend` (predicted customers on weekends), and `customer_ratio` (the ratio of daily predicted customers to the store's historical average). Feeding these highly relevant features into a single Global Random Forest model allowed it to learn overarching patterns across all stores at once.

## 5. Benchmark Models

As the baseline models are based on univariate statistical forecasting, it is assumed that historical sales data already captures the primary information required to predict future sales. Therefore, introducing external variables, such as promotions, holidays, is unnecessary here.

Since the benchmark models were designed as simple reference baselines, they were evaluated using a 42 day holdout test set instead of cross validation. This considerably reduced computational cost while providing a consistent baseline for comparison. Rolling validation was applied only to the Random Forest models.

### 5.1 Benchmark Model Selection

For this project, we selected Mean, Naive, Drift, Seasonal Naive, AutoARIMA, and ETS as our benchmark models. We want to set up a baseline ranging from simple to more advanced time series models. This helps us see what kind of forecasting performance we can get just by using each store's historical sales data. If Random Forest can't consistently beat these benchmarks, it means the extra complexity of feature engineering and machine learning might not actually be worth it.

We treat each store as an independent time series and train them on their own historical sales data. We used the last 42 days as the test period, and everything before that as the training set. All models use the exact same train/test split and are evaluated using primarily MAPE and MAE, ensuring a fair and consistent comparison across the board.

#### Why we use these models:

- Mean: It's simple, fast to compute, and provides a stable and computationally inexpensive long run level baseline, although it can still be influenced by extreme observations.
- Naive: The main perks are that it's straightforward, takes almost zero computing power, and carries over the most recent sales level.

- Drift: This builds a baseline with a simple trend, helping us see if adding a linear direction performs better than just using a flat average or the last data point.
- Seasonal Naive: Based on our EDA, we noticed strong weekly patterns. This model sets up a baseline using only weekly seasonality, letting us test how much just recognizing a 7 day cycle improves the forecast.
- AutoARIMA: This provides a more comprehensive, auto traditional time series benchmark that includes weekly seasonality. We use it to see if a more complex statistical model clearly outperforms simple rules of thumb.

```
=====
Mean and Median Accuracy by Model
=====
```

|                | MAE_Mean | MAE_Median | MAPE_Mean | MAPE_Median |
|----------------|----------|------------|-----------|-------------|
| model          |          |            |           |             |
| Mean           | 2126.523 | 2039.056   | 20.748    | 19.939      |
| AutoARIMA      | 2330.696 | 1744.770   | 31.556    | 21.150      |
| Seasonal Naive | 2146.890 | 2106.440   | 39.753    | 42.666      |
| Naive          | 5795.228 | 5588.250   | 97.778    | 100.000     |
| Drift          | 6025.591 | 5830.975   | 101.232   | 103.453     |

**Fig. 5 Table of Mean and Median Accuracy by Model**

After reviewing the initial results, we decided to add ETS as another benchmark model.

- ETS: As the results of metrics were not very ideal, and AutoARIMA also takes a long time to run searches across all stores, we brought in ETS to balance speed and accuracy. According to EDA Step 18 “Weekly and Month Sales Pattern”, our exploratory analysis showed a clear weekly pattern in the overall data, so we set the ETS seasonal period to 7 days. Even though the aggregated data doesn't show a glaring global trend, individual stores might still have their own local trends, so we kept the additive trend component and will validate whether it fits through our benchmark comparison. The actual configuration is set to additive trend, additive seasonality, and seasonal\_periods=7. All stores use the same ETS specification: additive trend, additive seasonality, and a seasonal period of seven days. But the model parameters and state components are estimated separately for each store.

```
=====
ETS Accuracy Summary
=====
```

|       | MAE_Mean | MAE_Median | MAPE_Mean | MAPE_Median |
|-------|----------|------------|-----------|-------------|
| model |          |            |           |             |
| ETS   | 1039.996 | 1003.901   | 17.071    | 16.304      |

**Fig. 6 Table of Mean and Median Accuracy for ETS**

## 5.2 Benchmark Model Results

Because the sales scale varies across different stores, a percentage error metric is much better suited for cross store comparisons. MAPE was calculated only for observations with non-zero actual sales. For the benchmark models, MAPE was first calculated separately for each store and subsequently averaged across stores.

### 5.2.1 Overall Results of the Initial Benchmark Models

Among the five initial models, the Mean model had the lowest average MAPE at 20.75%, making it the best performing benchmark overall before ETS was introduced. Even though Seasonal Naive's average MAE (2,146.89) was close to Mean's (2,126.52), its average MAPE reached 39.75%, which is significantly higher than the Mean model. This indicates that while Seasonal Naive's absolute errors might not be, but the relative error remains high when measured against each store's actual sales volume.

AutoARIMA delivered an average MAPE of 31.56%. While it beat Seasonal Naive, Naive, and Drift, it failed to outperform the simpler Mean model. This shows that even though AutoARIMA can automatically identify time dependent structures and factor in a weekly periodicity seasonality, its higher model complexity did not translate into a lower overall average forecasting error.

Naive and Drift performed significantly worse, showing that models that forecast by simply extending trends do not match our sales patterns. In contrast, ETS became the top performing benchmark model overall. Its average MAPE was 17.07%, lower than the previous best performer (Mean model at 20.75%). Its average MAE was 1,039.996, which was also noticeably lower than the Mean model's 2,126.523. The average MAPE of ETS is about 15% lower than that of the Mean model (in terms of relative reduction in Mean and Median of MAPE). Furthermore, the average MAE of ETS is about 1,086.53 sales units lower than the Mean model, representing roughly a 51.1% drop (in terms of relative reduction in Mean and Median of MAE). Both metrics provide that ETS's advantage in percentage calculation, but also reflects a genuine reduction in actual sales volume errors.

### 5.2.2 Results by Store Type

To verify whether the model only performs well on specific types of stores, we merged the benchmark results with the store\_type from the metadata, and calculated the average MAE and MAPE for store types a, b, c, and d separately. For each type, the model with the lowest average MAPE was chosen as the best model for that category.

The results show that ETS achieved the lowest average MAPE across all four store types. Therefore, ETS's overall advantage isn't driven by just one specific store type; it maintains the best average percentage error across categories a, b, c, and d.

Specifically, type b stores had the lowest ETS average MAPE at 11.93%, followed by type a at 16.52%, type d at 17.77%, and type c at 18.61%. However, there are only 11 stores in type b, far fewer than type a (377 stores), type c (75 stores), and type d (213 stores). As a result, its average results represent a much smaller sample size. Based on the current output, we can confirm that ETS ranks first across all types, but we cannot conclude that the differences between these types are statistically significant based on these averages alone. We identified the model with the lowest MAPE for each individual store, and then calculated the win rate for each model. The results are as follows:

| Model          | Stores Won | Win Rate (%) |
|----------------|------------|--------------|
| ETS            | 449        | 66.42        |
| Mean           | 135        | 19.97        |
| AutoARIMA      | 88         | 13.02        |
| Seasonal Naïve | 4          | 0.59         |

**Fig. 7 Ratio of Best Model**

A total of 676 stores were compared. ETS achieved the lowest MAPE in 449 of these stores, translating to a win rate of 66.42%, meaning ETS performed the best in roughly two-thirds of the stores. This outcome supports our overall average comparison: ETS is actually winning across the majority of individual stores.

## Summary

Considering the overall average performance, the results by store type, and the individual store win rates, ETS stands out as the strongest benchmark model for this project:

- ETS has the lowest overall average MAPE among all models.
- ETS has the lowest overall average MAE at 1,039.996.
- ETS is the model with the lowest average MAPE across all four store types.
- ETS came out on top in 449 out of 676 stores, achieving an individual store win rate of 66.42%.

We decide not to do more steps of cross validation. Not just for saving time, also because this evaluation strategy is consistent with the project objective of forecasting the next 6 weeks. Model robustness was assessed from multiple perspectives, including overall average MAPE, performance across all four store types, and the best performing model for each individual store. ETS consistently achieved the lowest average MAPE, ranked first across all store types.

Consequently, ETS should serve as the primary baseline for comparison with our upcoming Random Forest model. The Mean model and AutoARIMA still offer some reference value, as they achieved the best results in 19.97% and 13.02% of the stores, respectively. However, if we need to choose a single model to represent our overall benchmark performance, all evaluation metrics point to ETS.

## 5.3 ETS Forecast Result

We used ETS as our Final Benchmark Forecast Result and saved the sales predictions for each store into a CSV file for future use. The Total Sales predicted by ETS are shown below.



```

=====
ETS Final Forecast
=====
Stores forecasted      : 676
Forecast days         : 42
Total forecast rows    : 28392
Total sales result     : 173,830,086.84

```

**Fig. 8 ETS Sales Forecasting Result**

The results are consistent with the EDA finding that weekly seasonal structure is useful for forecasting. Moving forward, we should place a heavier emphasis on weekly cycles when making business decisions. Based on the findings from this timeframe, ETS is the most suitable model for us when we need to complete sales evaluations quickly due to time or cost constraints.

## **6. Random Forest Model**

### **6.1 Baseline Global Random Forest**

Before developing the final forecasting framework, a Baseline Global Random Forest was constructed as the starting point for model development. Unlike the statistical benchmark models presented before, this baseline serves as a reference for evaluating the contribution of subsequent modelling improvements.

The baseline model assumes that future sales can be explained using historical sales together with future known operational variables. Customer information is intentionally excluded to establish a reference model for subsequent comparison. It is assumed that stores share common sales patterns, allowing a single global model to generalise across different stores.

The model was trained on historical data from all stores simultaneously using historical sales lag features, rolling statistics, calendar variables, promotional and holiday information, and static store characteristics.

The baseline model followed the same training, hyperparameter tuning, and validation procedures as the final forecasting framework. The only difference was the exclusion of customer information. This model achieved a MAPE of 10.12%, providing a reference for assessing the effectiveness of the proposed two stage forecasting framework.

### **6.2 Customer-Augmented Forecasting Framework**

Although the Baseline Global Random Forest incorporated historical sales, calendar variables, and store characteristics, its forecasting accuracy indicated room for improvement. Exploratory analysis showed that customer traffic is strongly associated with sales, suggesting that customer information could further improve forecasting performance.

Since future customer counts are unavailable at the forecasting stage, a two stage forecasting framework was developed. This approach assumes that future customer demand can be forecast with sufficient accuracy and used as an explanatory variable for sales forecasting. First, a Global Random Forest model was trained to forecast customer counts

for the next 42 days using recursive forecasting. These predicted customer counts were then incorporated into the sales forecasting model together with future known variables, including promotional activities, store opening status, holiday information, and static store characteristics.

To capture customer counts more effectively, several customer features, including predicted customer counts (customer\_prediction), customer ratio, customer promo, and customer weekend, were included in the sales model. The proposed forecasting framework assumes that the historical relationship between customer counts and sales remains sufficiently stable over the 6-week forecasting horizon. During training, historical customer observations were used, while during future forecasting they were replaced by the customer forecasts generated in the first stage, preventing data leakage while preserving the predictive value of customer demand. It is assumed that recursively generated customer forecasts provide sufficiently accurate estimates for supporting downstream sales forecasting.

By incorporating customer forecasts, the average MAPE was reduced from 10.12% to 7.82%, demonstrating the effectiveness of the proposed two stage Global Random Forest framework.

| Stage       | Model                                    | MAPE (%) | Purpose                                           |
|-------------|------------------------------------------|----------|---------------------------------------------------|
| Baseline    | Global Random Forest                     | 10.12    | Baseline machine learning model                   |
| Enhancement | Global Random Forest + Customer Forecast | 7.82     | Evaluate the contribution of customer forecasting |

**Table 1. Comparison of Baseline and Enhancement Random Forest Model**

### 6.3 Hyperparameter Selection and Model Validation

To balance forecasting accuracy and model complexity, two Random Forest configurations were evaluated for both the customer and sales forecasting models. The candidate models differed in the number of trees, minimum leaf size, and the maximum number of features considered at each split.

All models were trained using the same feature engineering pipeline and evaluated using Rolling Forecast Origin Cross Validation. The forecasting horizon was fixed at 42 days, matching the required 6 week prediction period. This validation strategy preserves the chronological order of the data and prevents future information leakage. To improve computational efficiency, `refit = False` was adopted, meaning that the model was not retrained at every forecasting origin while maintaining a consistent evaluation framework.

Forecasting performance was evaluated using MAPE and MAE. MAPE served as the primary hyperparameter selection criterion because it enables comparisons across stores with different sales scales.

The table summarises the hyperparameter tuning results. For both forecasting tasks, RF200\_SQRT achieved the lowest MAPE and was therefore selected as the final configuration. Increasing the number of trees and the minimum leaf size did not improve forecasting accuracy. Consequently, both forecasting models adopted the same hyperparameter settings of 200 trees, `max_features = sqrt`, and `min_samples_leaf = 1`.

| Forecasting Task | Model            | Trees | Max Features | Min Leaf | MAPE (%)    | Selected |
|------------------|------------------|-------|--------------|----------|-------------|----------|
| Customer         | RF200_SQRT       | 200   | sqrt         | 1        | <b>8.63</b> | ✓        |
| Customer         | RF300_SQRT_Leaf5 | 300   | sqrt         | 5        | <b>8.77</b> |          |
| Sales            | RF200_SQRT       | 200   | sqrt         | 1        | <b>7.82</b> | ✓        |
| Sales            | RF300_SQRT_Leaf5 | 300   | sqrt         | 5        | <b>8.04</b> |          |

**Table 2. Model Selection**

## 7. Results

To evaluate the proposed forecasting framework, the final Global Random Forest model was compared with ETS, the best benchmark model identified .(MAPE = 17.07%).

Replacing ETS with a Global Random Forest reduced the MAPE to 10.12%. Incorporating predicted customer counts and customer features further reduced the MAPE to 7.82%.

Overall, the proposed framework reduced the average MAPE by 54.2% and the average MAE by 45.4% compared with ETS.

The improvement in forecasting accuracy can be attributed to two factors. First, the Global Random Forest captures complex nonlinear sales patterns more effectively than the statistical ETS model. Second, incorporating customer forecasts provides additional information about future demand, further improving forecasting accuracy. This is supported by the feature importance analysis, where customer variables rank among the most influential predictors.

| Model           | MAE      | MAPE         | Improvement        |
|-----------------|----------|--------------|--------------------|
| ETS             | 1039.996 | <b>17.07</b> | –                  |
| Final Global RF | 567.370  | <b>7.82</b>  | <b>↓54.2% MAPE</b> |

**Table 3. Comparison of ETS and Final Global Model**

## 8. Discussion

### 8.1 The Reason We Select Global Random Forest

A Global Random Forest was selected as the final forecasting model because it provides a good balance between forecasting accuracy and practical applicability. Compared with the baseline model, incorporating customer forecasts reduced the MAPE from 10.12% to 7.82%.

Unlike local models, the Global Random Forest is trained on data from all stores simultaneously, enabling it to learn shared sales patterns while requiring only a single model

to maintain. In addition, it captures nonlinear relationships among historical sales, customer forecasts, promotions, and store characteristics, while feature importance analysis provides useful business interpretability.

Overall, the Global Random Forest offers a robust and practical solution for large scale retail sales forecasting.

## 8.2 Complexity V.S Maintainability

The proposed forecasting framework balances model complexity and maintainability. Only a small set of Random Forest hyperparameters was tuned using Rolling Forecast Origin Cross Validation, keeping computational costs low while achieving strong forecasting performance.

The feature engineering process captures the main business drivers of sales, including temporal, customer related, promotional, and store features. Therefore, future maintenance is expected to focus mainly on periodic model retraining rather than extensive feature redesign.

Finally, maintaining a single Global Random Forest model is considerably simpler than managing separate models for individual stores, making the framework practical for long term deployment.

## AI Declaration

Used ChatGPT and Codex for code fixation and recommendation, better coding grammar way to approach the same goal and tidying up csv. production. We finally identified the result by ourselves.

Used ChatGPT for fixation and checking the indentation, consulting whether our solution is practical in real world for some steps and checking English grammar.

## Note for Code Execution

Since **AutoARIMA** takes a very long time to run for all stores, we prepared two versions of our notebook.py on moodle for convenience:

**Yu Ching Lin: The benchmark models are placed in the middle of the workflow.** This version presents the complete modeling process in chronological order and is intended to clearly demonstrate our full methodology.

**Xueying Peng: The benchmark section is moved to the end of the workflow.** This allows the Random Forest model to be executed directly without waiting for AutoARIMA to finish, making it much practical for reviewing the machine learning part.

Both notebooks produce the same final results and contain the same analyses; the only difference is the order of execution to reduce runtime during evaluation.